ModelIt!

Student Activity Pack

Can Plants Grow Without Soil?

What Plants Really Need to Survive

Grade 2 | NGSS 2-LS2-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think plants need to grow big and healthy?

2 Do you think a plant could grow if you just put it in water with no soil? Why or why not?

3 Draw a plant and label all the things you think it needs to survive.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Amount | Water Availability | Plant Growth

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Sun and Water

Sunlight high, Water high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: No Sunlight

Sunlight zero, Water high

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: No Water

Sunlight high, Water zero

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Nutrients in Water — Adding nutrients to the water to replace what soil usually provides — like plant vitamins

Air Flow — Moving air around the plant helps it get carbon dioxide, which it needs to make food in its leaves

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Do plants really need soil to grow, or can they grow with just water and sunlight? Use evidence from your simulation to support your answers.

1 Were you surprised that plants can grow without soil? What did the soil actually do for the plant?

2 If you were an astronaut, how would you grow food in space where there is no soil?

8 STEM Challenge: Build a Soil-Free Garden

ModelIt!

YOUR ENGINEERING MISSION

Design a small garden where plants grow without soil — using only water, sunlight, and nutrients.

THE SCENARIO

A space station needs to grow fresh food for astronauts, but there is no soil in space! Your team has been asked to design a small soil-free garden that grows plants using only water, light, and nutrients. Use what you learned to make your garden work.

Engineering Design Thinking

1. WHAT WILL YOU USE TO HOLD THE PLANT UP IF THERE IS NO SOIL — SPONGE, COTTON, OR SOMETHING ELSE?

2. HOW WILL THE ROOTS STAY IN THE WATER?

3. WHERE WILL YOU PUT YOUR GARDEN SO IT GETS ENOUGH LIGHT?

4. HOW WILL YOU ADD NUTRIENTS TO THE WATER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Agricultural Scientists study how to grow food better and help feed people around the world. Some work on hydroponic farms that grow food without soil. They earn $50,000–$90,000/year.